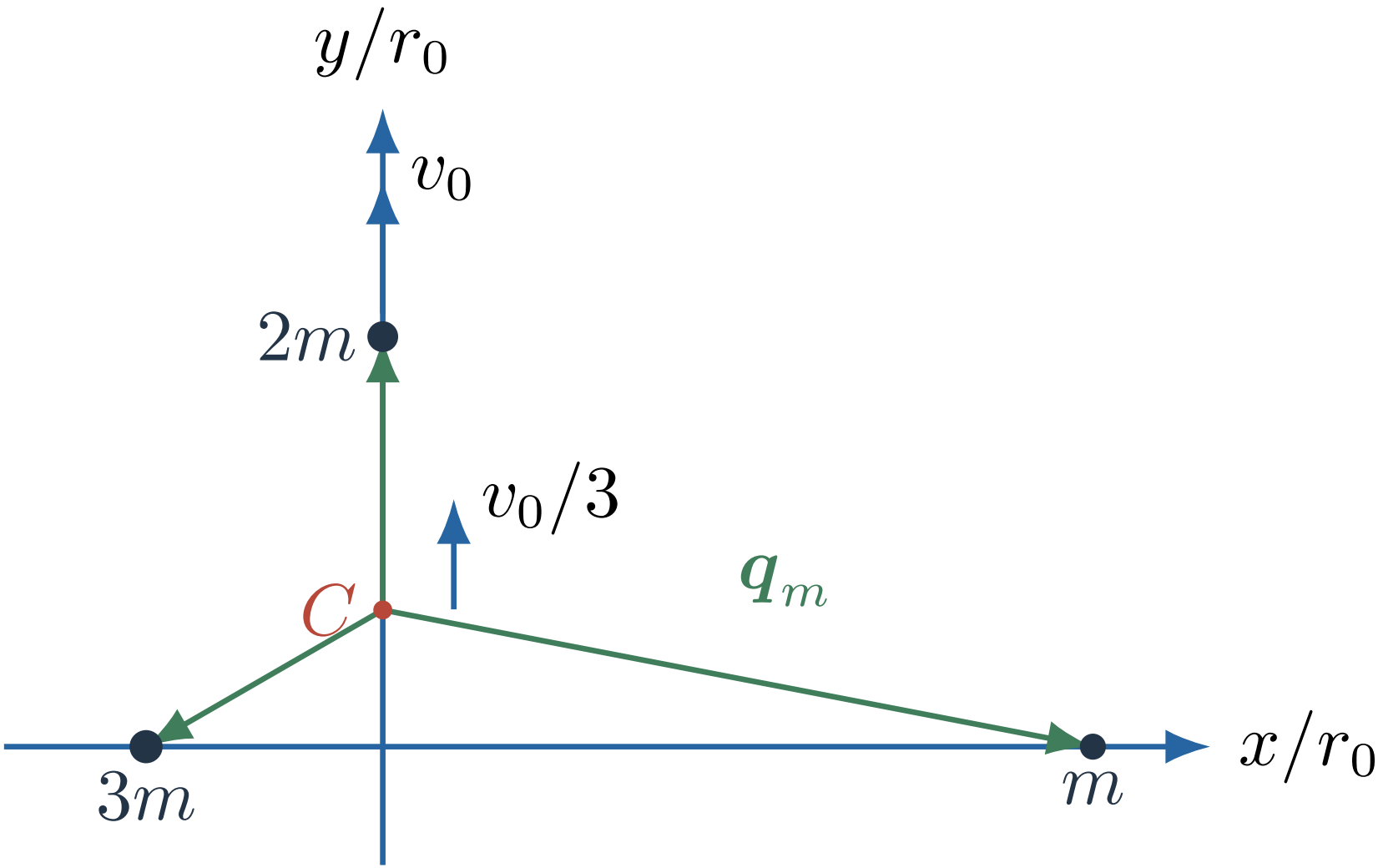

$$\omega_3 = \sqrt{6km}$$